Successful Moderation in Online Patient Communities

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Abstract

Background: Online patient communities are becoming more prevalent as a resource to help patients take control of their health. However, online patient communities experience challenges that require active moderation.

Objective: To identify challenges to having a thriving online patient community and moderation practices to manage an online patient community successfully.

Design: A qualitative study of Mayo Clinic Connect that draws insights from observations in community meetings, participating in the community, and semi-structured interviews with key stakeholders.

Results: We find five types of practices used to moderate the community successfully. These practices include instructive, connective, semantic, administrative, and policing practices. These practices lead to two sets of desired outcomes: community success and member success.

Conclusions: Successful moderation requires multiple practices to manage challenges in online patient communities – some from within the community's discussion threads and some from outside the community's discussion threads, some from authority figures and some from exemplary members. Different practices can influence different outcomes. Online patient communities can use these practices to effectuate desired outcomes.

Keywords: Online patient communities; moderation; guidelines; management

Introduction

Patients increasingly rely on the internet to look for medical information, ask questions, find peers with similar health concerns, read commentaries and experiences about health issues and consult reviews and rankings of treatments, doctors, and hospitals [1–4]. Seventy-two percent of internet users have looked at health information online within the past year. Online patient communities are becoming forums for patients to share their stories, gain peer support, and search for medical information [5,6]. Studies have shown that membership in patient communities is associated with better health, behavior, and medical knowledge [6–9].

To be successful, online patient communities must be vibrant. They should welcome members, encourage active participation and interactions, and provide accurate medical information. Among many concerns, the quality and accuracy of medical knowledge provided in these communities are questioned [10–13]. This is especially important given that members vary considerably in their medical expertise and may have difficulty discerning information quality. To address these issues and assure their success, some online patient communities use moderators [14]. Little academic research exists on effective moderation practices in online patient communities. Our objective is to systematically study the moderation practices in online patient communities and uncover contingencies leading to effective outcomes.

To gain a deep understanding of this phenomenon, we conduct an inductive field study of Mayo Clinic Connect (connect.mayoclinic.org), a leading online patient community. Using grounded theory methodology, we collect observational and interview data from key members over three years. Our findings highlight five moderation practices in the community: instructive, connective, semantic, administrative, and policing. Furthermore, these moderation practices are not only enacted by the community's management but also from enlisted volunteer community members who assist in achieving community success. Together, the practices used by both community management and community members successfully address several challenges to create a thriving community.

Prior Work

Online communities bring together members who share common interests and contribute valuable knowledge and expertise and are recognized as generators of significant knowledge [15]. In healthcare, online patient communities have become a medium for patients to share their stories, gain peer support, and search for medical information [5]. A healthy community retains its existing members, attracts new ones, and elicits their contribution [16]. Research has consistently shown the importance of feedback in driving members' contribution to online communities [17–19]. One challenge facing online communities is managing the sheer number of contributors and contributions in the absence of monetary incentives and formal managerial structures [20].

Compared to other online communities, online patient communities face some additional unique challenges. First and foremost, the quality, accuracy, and trustworthiness of medical knowledge provided over these platforms are questioned [10–13]. Second, there are ethical and privacy issues related to sharing and disseminating patients' information online [21]. Third, there is a mismatch of motives and expectations among members who use these platforms as communication and marketing tools and members who use them for social support and exchanging advice [22].

Community leadership plays an essential role in dealing with online community challenges [19,23,24]. Effective leaders and moderators are characterized by inclusivity, helpfulness and sociability [19,25,26].-Community Management can be assisted by elevating the status of exemplary members to become trusted volunteer peer leaders [18]. In online patient communities, the involvement of volunteer peers leaders and echoing the voice of patients are essential for community success [27].

Methodology

Research Context

We study Mayo Clinic Connect (referred to as Connect hereafter), a leading online patient community sponsored by Mayo Clinic. Connect is designed to "connect patients and family caregivers with each other" [28]. This community enables patients and their families with different medical backgrounds to share experiences, find support and exchange information with others who have faced similar experiences. The community has open boundaries, which means it is open to the public and not restricted to Mayo Clinic patients only. Currently, the community has 85,000 registered members and 10,000+ active members who have made a post within 30 days.

Data Collection

Initially, one author of this study participated in frequent informal discussions with the director of Connect and attended a community meeting discussing the concerns, successes, and goals of the community. During the meeting, the author observed the interactions between community management and peer leaders attending the meeting. The initial engagement with the community informed the theoretical sampling for the study and started the data collection. Additionally, all authors joined the community and observed members' contribution and interactions over the last three years.

The primary source of data came from semi-structured interviews with key stakeholders in the community. The first semi-structured interview was with the director of the community. This interview was instrumental in guiding subsequent interviews through a snowballing approach [29]. In two rounds of data collection, we interviewed the community director, the formally employed management staff, and fifteen other peer leaders. Through probing questions, we asked about their participation in Connect, and

what they felt the community did to achieve success. We asked them to expound on their experiences with the community – including challenging ones. Overall, we let the interviewees talk about being a member of Connect and their interactions with others. Each interview lasted approximately 30 minutes and was recorded and transcribed verbatim. Concluding each interview, we asked the interviewee to suggest other people to interview. We continued to interview key individuals until we reached theoretical saturation, and gained no additional insights from additional interviews [30].

Data analysis

We adopted a grounded theory coding perspective in order to let theory emerge from the data. We followed a constant comparative process in that throughout our analysis, we repeatedly compared codes and findings in frequent discussion among the authors [31].

Two of the authors conducted line-by-line open coding techniques [32] on the first five interviews. After both authors completed the coding of these first interviews, they discussed their findings to agree on emergent themes. One author coded the remaining interviews and all authors met frequently during the coding process to discuss the inductive insights from the codes. After creating an extensive list of open codes, we conducted a second round of coding to ensure that minor differences in codes from open coding were combined to form unique ideas that did not overlap in the data. During open coding, memos were kept on each interview to provide extended notes and insights.

After completing open coding and discussing emergent themes, we followed an axial coding process to combine and relate codes together to form more abstract themes and relationships [33]. Axial coding helps to identify the who, what, when, where, why, and how of the emerging theory. In our case, it illustrated a clear picture of the moderation practices in the community.

Results

Community Challenges

Our interviews highlighted certain challenges that require active moderation practices to address. First, many members are passionate about Connect and their contributions to it. These members experienced or overcame medical conditions and are passionate to share their experiences and help others. Thus, passion can yield positive consequences, such as increased engagement and participation. Passion, however, leads many individuals to argue about different beliefs and opinions. Ugly and argumentative posts are at times a challenge for Connect. *"We might have a member that comes in really opinionated." – Moderator 3*

Second, members in Connect are not required to disclose their real identities. These socially ambiguous identities allow individuals to share experiences without worrying

about judgment or stereotyping and to alleviate privacy concerns. Yet, negative consequences come when identities are pseudonymous because anonymity creates an environment with no consequences. This increases the chance of social inhibition, and therefore, the expression of anti-social behavior such as bullying, arguing, and trolling may emerge. *"Right, there're no names involved. It's just the member's handle and you can disclose or say what you want." – Mentor 8*

Third, sharing personal experiences is perhaps the overarching activity of members in Connect. Members with similar problems share experiences to encourage, inform, and support one another. Overall, this activity creates positive outcomes, and many members are very grateful for the feeling that they are "not alone." However, challenges arise when experiences are viewed as or claimed to be "medical advice". There is the potential for legal repercussions if that advice fails to provide positive outcomes or causes further harm. *"When it comes to alternative medicine, complementary medicine, that's where we run into problems. Because members are free to express their views, but when they advise their discussion group and they instruct them to stop taking traditional medications and go for homeopathy or herbal things, that's when we have to step in." – Moderator 4*

Fourth, soliciting personal information is helpful for community members to know how to respond to other members. More information regarding a member's condition and situation help others relate to the member's issues and provide appropriate support. Members often ask questions such as "how was this diagnosed", "how long have you had the problem", and "what kinds of treatment have you had." However, sometimes this is taken too far. Violations to privacy may occur when a member shares excessive personal health information. *"In a HIPAA sense, this individual is asking this person very specific questions on the day, the time and the results of a medical test, and I'm not sure that that's an appropriate place for a public discussion board." – Mentor 6*

Moderation of the Community

These challenges are addressed by community management. The management team of Connect consists of a director and five staffers. These managers are referred to as moderators in the nomenclature of Connect. Moderators have set formal guidelines such as being careful when giving out medical advice, being respectful, and no commercial advertising. At the outset of the community, the moderators ran the community, moderated others' posting, and provided the support to members that started to join the community.

As the community grew, moderators recognized the need to scale community moderation to maintain members' contribution. At the same time, moderators noted that some members were taking increasingly active leadership roles. They identified these emerging peer leaders and invited some of them to help monitor the community. In the nomenclature of Connect, the emerging peer leaders selected by the moderators are called mentors. Once selected, mentors, who volunteer their time, are trained and guided by moderators to act in

accordance with the guidelines of the community. Multiple training practices are implemented in order to onboard and train mentors (Table 1).

Table 1. Training mechanisms for mentors.

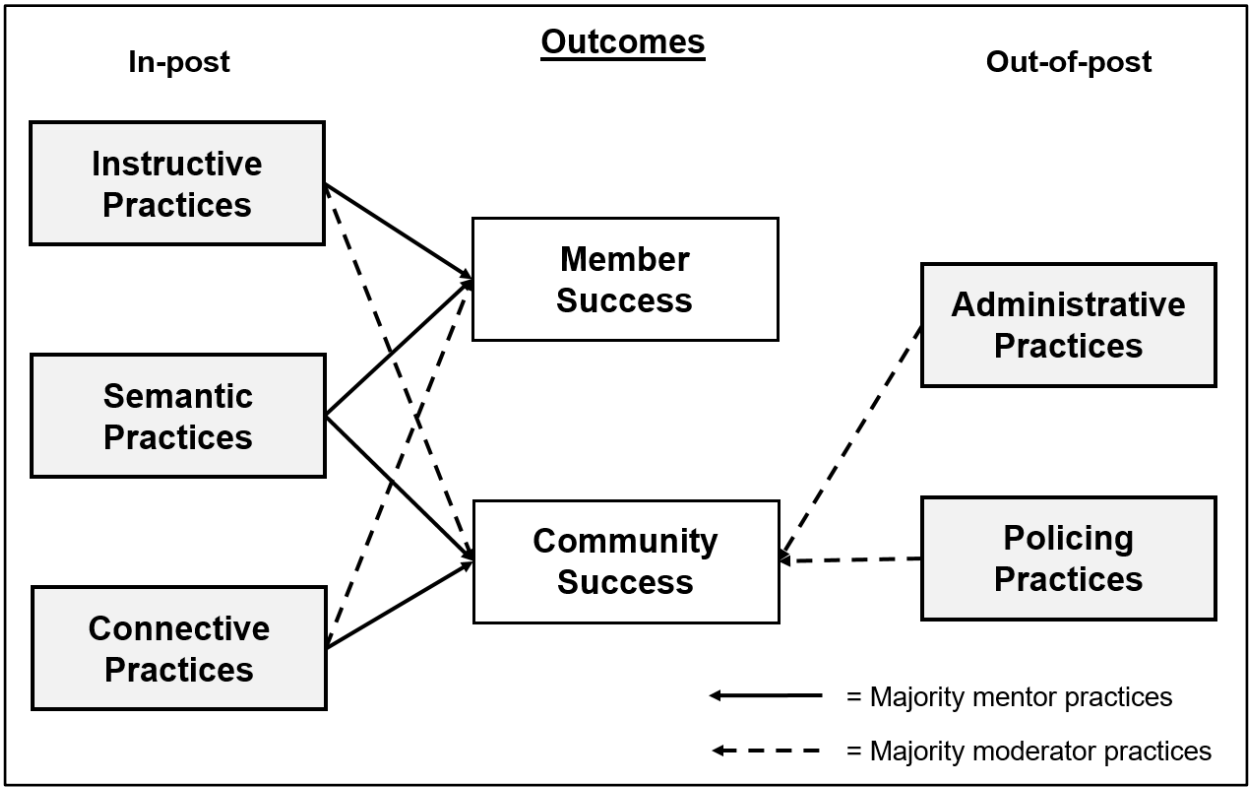
Training Mechanism	**Example from Interviews**
Standard documentation	*"I'll send you three or four documents that you can look through and understand a little more what we're looking for in a mentor."*
Setting expectations	*"We're hopeful that you'll agree to spend at least X number of hours a month visiting the board and watching things."*
Coaching/Individual Training	*"When you are first invited, you spend time with the director, a one-to-one, doing screen shares and discussing different scenarios."*
Coaching/Group Training	*Moderators set up a private "mentors & moderators" group on Connect*
Mentor Meetups	"*The mentors and moderators, we get together once a year. It is a little bit more training and sort of continuing education of being a mentor."*
Remote Training	"*Every quarter we will have a phone call where we can share what concerns us and build up a discussion."*
Behavior Emulation	"*We [Moderators] ask them to emulate us as far as the way we would respond"*

These training practices communicate and transfer the goals of moderators to mentors. This process brings three advantages. First, it reduces the workload of moderators, which is becoming infeasible with the continuous growth of the community, receiving around 10,000 posts per month. Second, mentors are uniquely qualified to assist members because of their shared experience – an experience that the moderators often do not have. Third, it enables moderators and mentors to specialize in different tasks focusing on desired outcomes. Moderators focus on resolving tensions, administrative responsibilities, and setting up the agenda to improve community success and growth. Mentors can focus on patient success and support within the daily operation of the community.

Desired Outcomes and Practices

Two categories of desired outcomes emerged from the data: community outcomes and member outcomes. Community outcomes describe the overall success of Connect and include positive culture, active engagement, and adherence to guidelines. Member outcomes focus on a member's well-being, support, and reactivation. We found that moderators and mentors enacted different practices to achieve different outcomes. Figure 1 highlights how practices and outcomes are connected.

Figure 1. Moderation Practice Groups and Desired Outcomes



In Figure 1, the relationships between moderation practices and outcomes are the result of axial coding. Outcomes are displayed in the center with practices on the sides. Some practices occurred inside the community discussion threads (in-post), and some occurred outside discussion threads (out-of-post). Groups of similar practices were abstracted into high-level categories that added parsimony to our understanding of the relationships. These practices were not shared equally between moderators and mentors. We display the practice-outcome relationship that the majority of moderators implement in with a dashed line, and the relationship that the majority of mentors implement with a solid line. This figure shows how moderators and mentors play different roles in addressing challenges of the community.

The figure shows five practice groups. Instructive, semantic, connective, administrative, and policing. *Instructive practices* are actions that provide medical information relevant to member experiences. *Semantic practices* are actions that focus on the content of messages and responses; that is, what is being communicated inside a post. *Connective practices* are actions that create relationships inside the community and bring individual members into a community setting. *Administrative practices* are performed by formal management outside of discussion threads to keep the community growing with a "positive community feel." *Policing practices* are interventions against anti-social behaviors and departures from community guidelines. Table 2 details specific practices that make up the practice groups

and then the specific aspect of member or community outcomes that are impacted. Figure 2 illustrates some of these practices in an example post from Connect.

Table 2. Detailed practices with definitions and impacts

Practice Category	Practice with definition	Outcome impacted
In-post practices		
Instructive	**Pointing to doctor:** Instructing a member with a medical condition to see a doctor	**Member:** Member well-being Community: Adherence to guidelines
	Providing information: Searching and sharing online information to help manage health conditions and follow community guidelines	**Member:** Member well-being Community: Adherence to guidelines
	Bringing in medical help: Involving medical professionals from Mayo Clinic to provide answers to medical questions and concerns in posts or through live events	**Member:** Member well-being
Semantic	**Validating Concerns:** Relating empathetically to what a member is going through	**Member:** Member support
	Sharing experiences: Including past experiences inside posts	**Member:** Member well-being Member support **Community:** Active participation
	Welcoming members: Providing words of welcome when a new member joins the community	**Member:** Member support **Community:** Active participation Positive culture
	Asking questions: Ending a post with a question to promote continued discussion	**Community:** Active participation
	Positive reinforcement: Including constructive and uplifting language in posts	**Member:** Member support **Community:** Positive culture
Connective	**Tagging:** Using the @ symbol to link a member to a specific post and send them a notification	**Member:** Member reactivation **Community:** Active participation
	Bringing from private to public: Taking conversations initiated offline and inserting them into the public discussion thread	**Community:** Active participation
	Proactive follow-up: Reaching out to members who have not participated recently	**Member:** Member reactivation
Policing	**Asking for evidence:** Validating claims from members by asking for scientific backing	**Community:** Adherence to guidelines

	Diffusing criticism: Addressing antisocial behavior and turning into positive behavior	
Out-of-post practices		
Policing*	**Reporting posts:** Monitoring the community for deviance to the guidelines and reporting it to moderators	**Community:** Adherence to guidelines
	Muting members: Disabling the ability for a member to post content	
	Taking down posts: Removing posts from the community that violate the terms of use.	
Administrative	**Increase mentors:** Recruiting exemplary peer leaders and inviting them to be mentors	**Community:** Positive culture
	Coaching: Working with difficult or new members one-on-one offline to improve behavior	
	Spotlighting members: A specific program to share an individual member's information with the community	
*Note: Policing practices contained actions that occurred both in-post and out-of-post		

Figure 2. Moderation practices employed in one example post

Posted by @new_member

I was diagnosed with ***** about a year ago. I WAS a very active ***** year old ***** all of a sudden, not being able to get out of bed on her own and having severe ***** pain. As is customary, after waiting 3 months to see a ***** doctor, she administered the ***** test and I was VERY high. So, she diagnosed me with ***** and prescribed ***** mg of ***** . I don't like taking meds (as most of us don't), but I was in pain and had no quality of life. I'm a swimmer and could hardly dog paddle ! One year later, I have been weaning off the ***** mg down to ***** mg. My ***** said that now it's a matter of determining if I have ***** or ***** … so, she prescribed ***** mg of ***** for a month. I will say my ***** pain seems better, but my ***** are awful !! I don't want to go back on ***** – the devil drug. I would like to get to the "crux" of what is CAUSING my ***** ??? No doctors seem to address that? What would be my next step? A nutritionist? I am ***** of taking supplements, as well. Thank you for any help or sharing your experience. Weird disease … I just want my old life back – being able to move without pain, etc. Too much to ask?

Posted by @mentor

Hello @new_member — welcome to Mayo Connect. We are glad you found us. Connect is a great place to ask questions and learn what others with similar health concerns are doing for treatments. I also have ***** . As my ***** explained it to me ***** is like ***** all over the body. My first episode of ***** was in ***** and I was started on ***** mg ***** . It took me ***** years to taper off of ***** and the ***** went into remission and stayed gone for ***** years, It came back in ***** and again I was started on ***** mg of ***** . I'm now controlling it with ***** mg dosage and hoping to be off of the nasty stuff in a month or so but there are no promises. The one thing I can tell you is we are all different so you just need to take it one day at a time and keep moving but not over doing it.	**Welcoming members** **Validating concerns** **Sharing experiences**
You have asked the ***** question – what is causing the ***** ? Unfortunately I think the answer is the disease itself, but I'm no doctor and have no medical training. Mayo Clinic has a good overview of ***** here: https://www.mayoclinic.org/diseases-conditions/*****	**Providing information**
I'm tagging other members who have discussed ***** and ***** to see if they can offer you any suggestions or more information. @member1, @member2, @member3, @member4 can you offer any suggestions or information for @new_member ?	**Tagging** **Asking questions**
I would also like to mention that I think you are on the right track with nutrition. You might want to check out a book by Dr. ***** – The ***** Protocol. She has ***** and has a great story about using nutrition for ***** to reduce or eliminate her ***** symptoms. You can read more about her here: https://external-site.com	**Providing information**

Posted by @moderator

Hello @new_member, I'd like to add my welcome, and thank you for joining the Connect community, and sharing your concerns. @mentor has offered some great insight, and while we wait for members to add to the conversation, I thought you might wish to view these step-by-step instructions to help you get the most out of Connect. Here's the link to the webpage: https://connect.mayoclinic.org/get-started-on-connect/	**Welcoming members**
If you need additional help, you can also contact a Connect community moderator by filling out this form: https://connect.mayoclinic.org/contact-a-community-moderator/	**Providing information**

Achieving Desired Community Outcomes

First, having a positive culture was a core theme in the conducted interviews. While moderators attempt offline administrative programs such as get-to-know-you campaigns and coaching members to create a positive culture, mentors contribute more to culture through semantic practices inside posts. Ultimately, positive culture is achieved through members exemplifying mentors' behavior.

Second, active engagement in the community is the proactive and consistent participation from members in discussions. Engagement is achieved through two main practice categories: semantic and connective. While mentors implement the majority of semantic practices, both moderators and mentors use connective practices. One explanation for why moderators continue to focus on connecting members is because they have participated in the community longer, and therefore have more knowledge of potential relationships. In contrast, semantic practices do not require knowledge of relationships. Semantic practices only require taking an interest in new posts from members.

Third, adherence to guidelines entails the community not deviating from posted guidelines – such as remaining "safe, supportive, inclusive, and respectful." In interviews, the guideline referenced most was caution in giving out medical advice. The community needs to ensure that no member posts a personal solution to a diagnosis without stating that it is only personal experience. Additionally, the community watches for anti-social behaviors such as bullying, obscene language, and personal or commercial advertisements. These transgressions are outlined in the Terms of Use. Such harmful behaviors are addressed mainly by moderators using policing practices. Mentors do not participate directly with interventive policing practices, instead, they act in a reporting role. Table 3 lists the three community success outcomes with associated excerpts.

Table 3. Practices used for community success outcomes

Community Success Outcome	**Practice Group**	**Excerpt**
Positive Culture	Semantic	*When I see a new person come into the Connect community, I have a tendency to welcome them and try and establish some kind of early sense of the fact that we're a community, not just a cold chat board. - Mentor 6*
	Administrative	*And with a lot of coaching, that particular person, over the course of many months, ended up becoming a really valuable member. And rather than talking down to members, now currently actually really encourages members – Moderator 2*
Active Engagement	Semantic	*Asking questions of the person who is posting is a good thing. I know that people don't always like to post on a public forum, even if it's anonymous. I always start a sentence with if you are comfortable sharing more information about yourself, would*

		you please tell us how this was diagnosed and what kind of treatments you have experienced, et cetera? - Mentor 7
	Connective	*I try to connect them. Being a moderator of the group, you learn the conversations that, you're familiar with, so sometimes, somebody will post something, and I'll go, "Oh! She talked about this, too!" - Moderator 3*
Adherence to Guidelines	Policing	*It's being there to intervene if people post their personal contact information, which could put them at some sort of privacy risk or they're writing in that they feel like they're close to ending their life or this kind of thing. - Moderator 1*
	Instructive	*We're just other members in a community who have had similar experiences but you need to find a doctor. The one thing that I will recommend again and again is that it's really critical for patients to go to a doctor who's an expert in their condition - Mentor13*

Achieving Desired Member Outcomes

In addition to achieving community success, the overarching goal for Connect is member success. The first member success outcome expressed by community leaders is member well-being or a members' positive and safe progress with their health. This outcome is achieved through two practice groups: semantic and instructive. Most of these practices come from mentors because of their similar experiences. Moderators, however, have the authority to speak to doctors and get feedback. Additionally, moderators may host webinars with doctors who answer common questions that have been raised by the community. These webinars are the only out-of-post practice for member well-being, where most of the practices to improve member well-being come from in-post.

Second, member support is realized when a member does not feel alone, has appropriate resources to meet his/her needs, and has the emotional support to face his/her condition. In-post semantic practices provide the most support for this outcome. Specifically, welcoming members, sharing personal experiences, and validating concerns was the content that supported patients. There is evidence of moderators using these semantic practices, yet moderators most often use them solely to give the mentors an example to emulate or in the absence of a mentor with a shared experience. More recently, mentors are the central role focusing on this outcome due to the mentor's shared experience and the ability to relate to patients. An excerpt that highlights the different roles of moderator and mentor is a moderator who talked about supporting a patient but clearly stated in the interview that she "took off her moderator hat" and acted as a relatable mentor for that specific post.

Third, individual patient reactivation is a significant focus for Connect. Many of the discussion groups revolve around chronic conditions, and therefore when members drop off from the discussion, the community is not able to support that member. In order to achieve patient reactivation, moderators and mentors tag inactive members and proactively follow up to see how they are doing and invite them to be an active participant.

Even though this appears to be more of a mentor task, moderators often follow up because of their existing relationships. Table 4 lists the three member success outcomes with associated excerpts.

Table 4. Practices used for member success outcomes

Member Success Outcome	**Practice Group**	**Excerpt**
Member well-being	Semantic	*What we're most anxious to do is encourage them to take on more of the responsibility managing their own circumstance and dealing with their medical team. - Mentor 4*
	Instructive	*It's a live video Q&A session between a pediatric physician and the surgeon… we post that on Facebook and then we also have it linked to Mayo Clinic Connect, so the members there are able to watch that video live, ask questions before and during the live segment takes place. - Moderator 3*
Member support	Semantic	*I don't very often take off my moderator hat and speak from a personal perspective, but someone wrote in about a situation they had and I had quite a similar one, and no one else was really responding, so I could provide some value and benefit to this member, really by kind of reinforcing that what she had was a valid problem, and explaining where things went in my situation. - Moderator 1*
	Semantic	*Yeah, so you have your peer-to-peer support, and people who have had the same experiences as you. - Mentor 11*
Member Reactivation	Connective	*I would go back and find people that I knew had been on before and weren't well and we just hadn't heard from them in a while. I would check back and ask them how they're doing. - Mentor 7*
	Connective	*If a member has been posting and then suddenly stops, we tag members to connect them to our members. - Moderator 4*

Conclusion

Like most online communities, Connect experiences multiple challenges. We have identified and explicated many practices used by Connect moderators and mentors to achieve the desired outcomes of community success and member success. These practices are not solely emergent nor wholly prescribed. The nature of the practice matches the needs of the community and its members. Some of these practices are *interventive*. They are mostly used to provide a quick and direct resolution to a time-sensitive challenges. Other practices are *preventive*. They promote members' behavior that leads to desired outcomes by avoiding tensions. These practices can be a guide and reference for an online patient community to promote the success and well-being of its members.

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Conflict of Interest Statement

The authors have no conflicts of interest to declare.

Data Sharing Statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request and approval of IRB.

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